a. Detailed L_2-Convergence Check

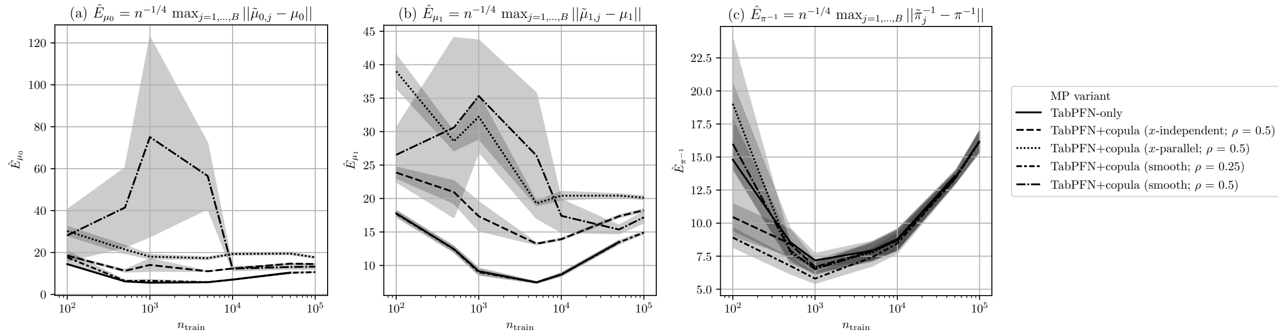

Figure 1. L_2-convergence check for every nuisance function based on the synthetic data with varying size of the train data, n_{train} (here $d_x = 25$). Reported: mean L_2 error $\hat{E}_\diamond \pm$ se over 10 runs (lower is better), where $\diamond$ corresponds to different nuisance functions: (a) $\diamond = \mu_0$, (b) $\diamond = \mu_1$, and (c) $\diamond = \pi^{-1}$. **Results.** These results indicate that performance improves with sample size up to a moderate regime, consistent with the L_2-convergence condition required for the BvM theorem. *However, beyond this regime, the concentration of the inverse propensity score posterior deteriorates, which reflects a difficulty of accurately recovering inverse propensity scores for propensities close to zero and one with TabPFN*. Together, this confirms that, in practice, the contraction rates are sufficient to yield stable and well-aligned uncertainty estimates, yet only for medium-sized datasets.

b. Memory Usage & Runtime

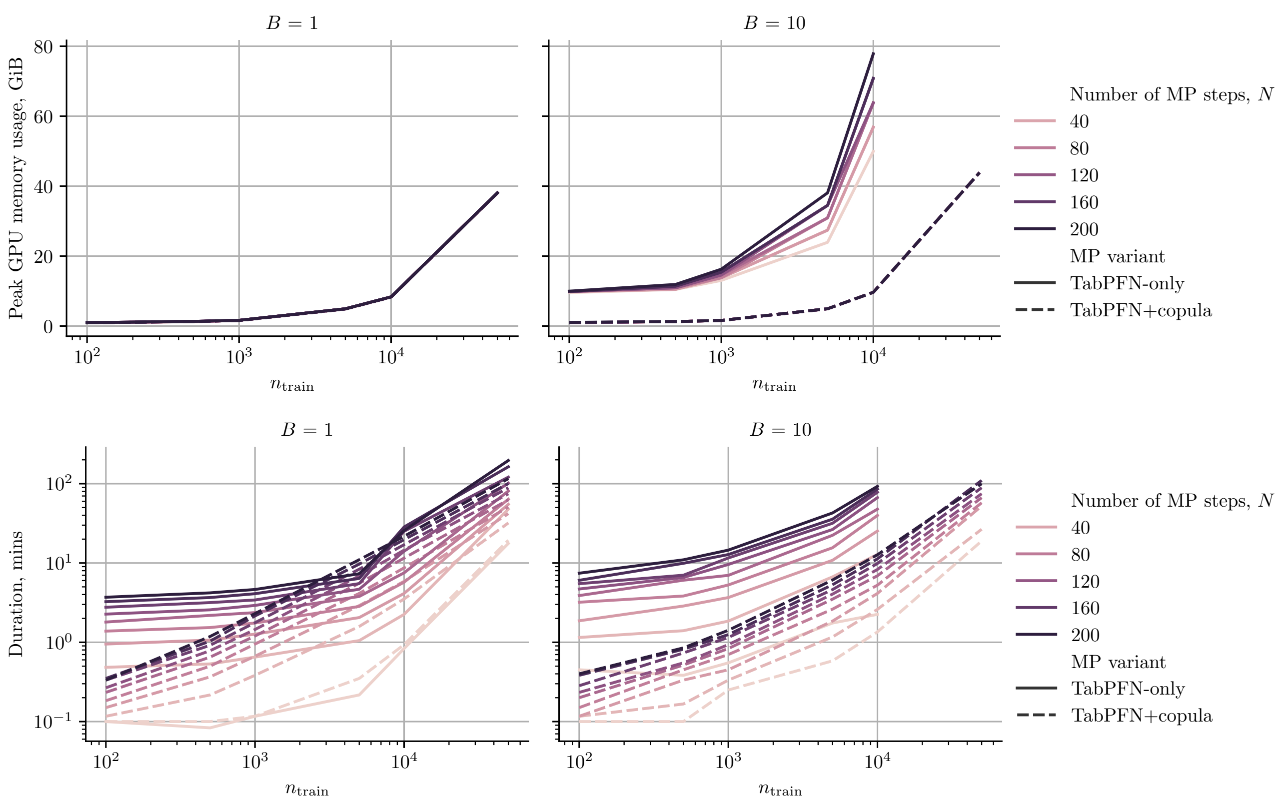

Figure 2. Peak GPU memory usage (top figure, in gigibytes (GiB)) and total runtime (bottom figure, in minutes) of ATE experiments for our *MP-OSPC* with different MP variants and varying number of MP steps based on the synthetic data with varying size of the train data, n_{train} (here $d_x = 25$). Experiments are carried out on AMD EPYC 9455 48-Core CPU and 1 H200 Nvidia GPU. **Results.** While a TabPFN-only MPs are very GPU-memory/runtime intensive for increasing number of posterior draws B (as every draw has to be parallelized and requires N subsequent TabPFN calls each with up to $O((n + N)^2)$ time complexity), the copula-based MPs are more GPU-memory/runtime effective (as they require one initial TabPFN call with $O(n^2)$ time complexity and all the MP inference for B draws can be easily vectorized).

c. MP convergence

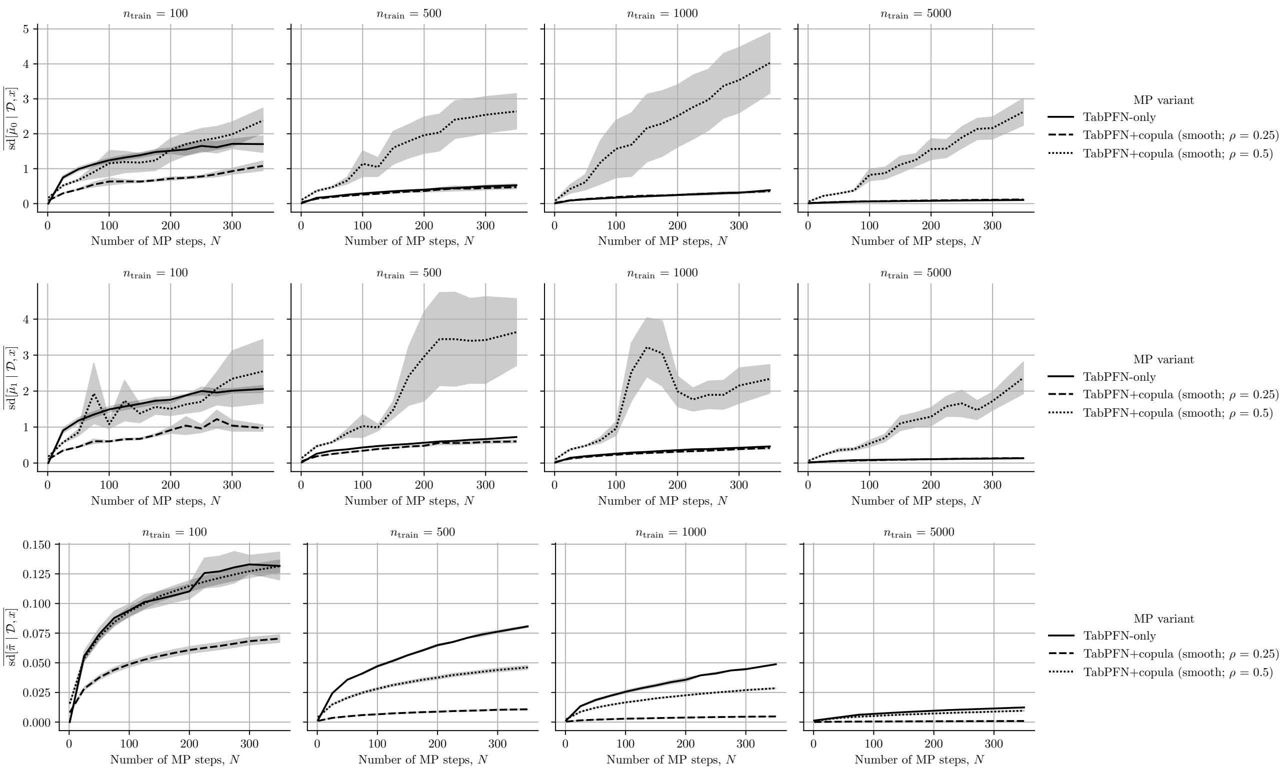

Figure 3. Convergence of different MP variants wrt. increasing number of MP steps N for every nuisance function based on the synthetic data with varying size of the train data, n_{train} (here $d_x = 25$). Reported: mean posterior standard deviation $\overline{\text{sd}[\tilde{\diamond} \mid \mathcal{D}, x]} \pm$ se over 10 runs, where $\diamond$ corresponds to different nuisance functions: (top row) $\diamond = \mu_0$, (middle row) $\diamond = \mu_1$, and (bottom row) $\diamond = \pi^{-1}$. **Results.** We see that, with few exceptions for large ρ, the posterior standard deviation generally flattens with the growing number of MP steps (e.g., for $N = 300$ with $n = 100$, for $N = 250$ with $n = 500$, etc.)

d. Sensitivity Checks Synthetic Dataset

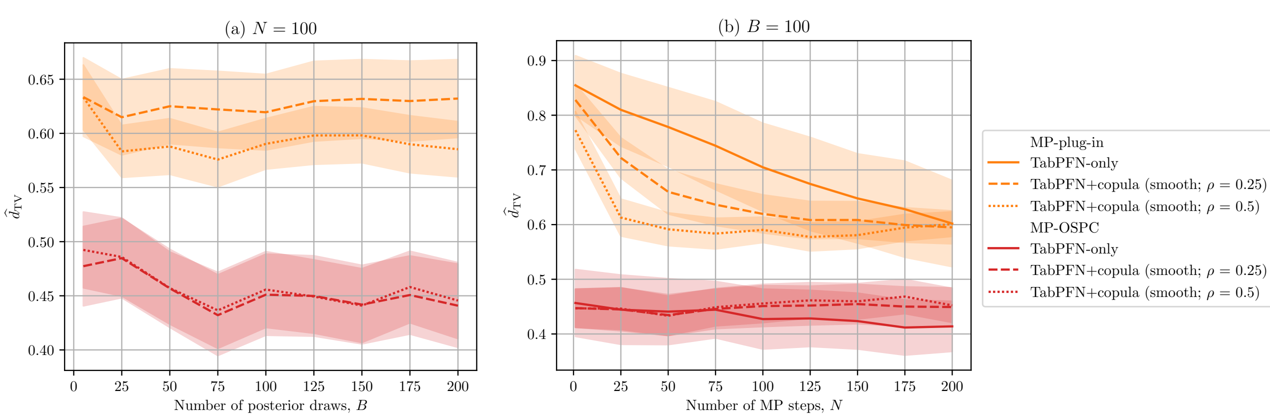

Figure 4. Quality of the asymptotic uncertainty for Bayesian ATE estimators based on the synthetic data with (a) varying size of posterior draws, B, and (b) varying number of MP steps, N (here $n_{\text{train}} = 500, d_x = 25$). Reported: mean $\hat{d}_{\text{TV}} \pm$ se over 40 runs (lower is better). **Results.** Our MP-based ATE posteriors converge already when $B = 75$ and $N = 100$. Importantly, the performance of our MP-OSPC remains relatively stable under different hyperparameters highlighting the first-order insensitivity to the misspecification of the nuisance functions posteriors.

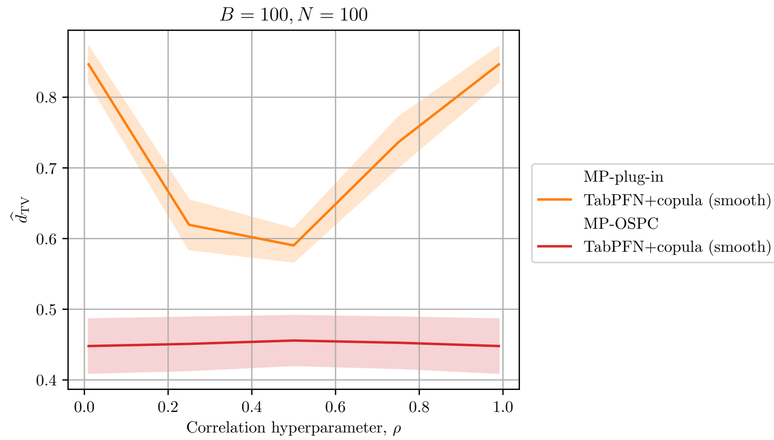

Figure 5. Quality of the asymptotic uncertainty for Bayesian ATE estimators based on the synthetic data with varying correlation hyperparameter, ρ ($n_{\text{train}} = 500, d_x = 25$). Reported: mean $\hat{d}_{\mathrm{TV}} \pm$ se over 40 runs (lower is better). **Results.** While the performance of the MP-plug-ins heavily depends on the correlation hyperparameter ρ, the performance of our MP-OSPC remains stable highlighting the first-order insensitivity to the misspecification of the nuisance functions posteriors.